

A Descriptive study to assess Depression, Anxiety & Stress among higher secondary students of Government schools of Chandigarh, India

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Abstract: A cross sectional descriptive study was done to assess depression, anxiety and stress among 288 higher secondary students of government schools of Chandigarh, India using self-administered structured questionnaire. Data was entered and analyzed using SPSS version 23. **Objectives:** To assess depression, anxiety and stress among students of higher secondary school and to determine the association between depression, anxiety and stress with socio demographic variables. **Results:** A majority of students, 29.9% had moderate depression, 18.8% mild depression, 13.9% severe depression and only 2.4 % had extremely severe depression. The mean score of depression was 13.05 ± 7.2 . The majority of students, 26.5% had moderate anxiety, 25.3% had severe anxiety and 21.5% had extremely severe anxiety. The mean score of anxiety came out to be 14.09 ± 6.7 . Majority of students 44.8% had moderate stress, 21.5% had mild stress and 8.3% had severe stress. The mean score for stress was 15.6 ± 7.1 . Association between anxiety and depression ($p < 0.001$), Odd's Ratio (OR)= 10.93, between stress & anxiety ($p < 0.001$), OR=9.771, and between stress and depression ($p < 0.001$), OR=17.2 came out to be highly significant results in association. **Conclusion:** The variation in level of depression, anxiety and stress was found to be present high among higher secondary school which showed higher percentage of anxiety in comparison to depression and stress. Higher anxiety and depression percentage could affect the individual mental health both in short and long run which could hinder the academic growth as well as personal growth of students.

INTRODUCTION

The World Health Organization defines adolescence as “the period in human growth and development that occurs after childhood and before adulthood” and considers this to be the period between the ages of 10 years to 19 years of age. Adolescents [aged from 10 to 19] account for approximately 17% of the world's population¹.

Adolescence is an essential human developmental stage, during which biological, psychological, and social development occur. Such changes and challenges during this developmental stage can result in mental health issues, such as depression and anxiety, which are common in adolescence and have been found to have profound effects throughout adulthood. Depression and anxiety have been recognized by the WHO as two major mental health problems, which usually begin during adolescence. Therefore, it is important to identify the reasons behind these mental health issues among adolescents for further interventions. One critical developmental experience for most adolescents that may influence their mental health for adolescents in primary and secondary schools, academic related events are believed to be major stressors, especially in Asian countries such as China, Korea, and Singapore.

Adolescents are always considered to be full of energy and bright hope of the future. The stage of adolescence is characterized by significant physical, emotional and intellectual changes, and changes in social roles, relationships and expectations and is marked by stress¹. During this stage, some of adolescents, experience anxious feeling and overwhelming sense of fear while others may have lack of confidence and feeling of unwellness. Anxiety/depression symptoms/disorders in adolescents result in significant impairment in social and academic functioning; produces a substantial distress for both students and family, apart from teachers². Moreover if not treated, these symptoms tend to persist, and increase the risk for medical illnesses and various psychiatric disorders particularly depression and substance abuse³.

Stress related disorders in adolescents are thus important issues that draw researchers' attention since it is related to numerous problems. Previous studies revealed that anxiety is associated with substantial negative effects on children's social, emotional and academic functioning⁴, poor social and coping skills⁵; loneliness, low self-esteem⁶, school avoidance, decreased problem-solving abilities, and poor academic achievement⁷. Earlier research reported that adolescents are at highest risk for onset of social anxiety disorder or social phobia⁸. In a review of the main studies carried out in the general population it was concluded that there had been a one standard deviation increase in rates of anxiety in children, adolescents, and young adults⁹. Depression has serious impact on every person, adolescents are not any exemption. This can lower the quality of life and wellbeing, increases the risks of suicide and can worsen mental or physical health. It is well proven by research that depression in adolescent is highly associated with increased risk of suicidal behavior,

homicidal ideation, tobacco use and other substance abuse into adulthood¹⁰. The majority of suicides in India are by those below the age of 30 years and also that around 90% of those who die by suicide have a mental disorder¹¹. About 8% of children and adolescents suffer from depression and 11% of adolescents have a depressive disorder by the age of 18 years according to the National Co-morbidity Survey- Adolescent Supplement Research on Psychiatric disorder among school samples of adolescents was found in about 29% of girls and 23% of boys with depression being the most common disorder. In addition to anxiety and depression adolescents are more prone to stress. Stress may take many forms like academic stress, peer pressure, relationships, negative emotions and low economic status of family etc. Achievement anxiety is one of the most common causes of stress in students. There is an increasing concern regarding stress and its relationships with mental health problems among school going adolescents in India.

Today's school life develops anxieties, tensions, frustration and anger leading to a negative impact on the mental health of students. As a result more and more students are facing adjustment problems in every sphere of life- personal, social and emotional. The World Health Organization (WHO) reported in 2001 that about 450 million people worldwide suffer from some form of mental disorder and that one in four people meet criteria at some point in their life.¹ Depression is a major cause of morbidity, ²with a lifetime prevalence varying from 3% in Japan to 17% in the US. In most countries, the number of people who would suffer from depression during their lives falls within an 8-12% range. At any given point of time, 25% students reported symptoms of Depression.⁴ In India 90% of people who need mental health support do not have access to it. There is a significant human resource gap of mental health professionals.

Only a few studies on Indian adolescents are available. They show that adolescents face multiple stressors such as criticism from parents, teachers, and peers; interpersonal problems; problems in living conditions and home environment; worries about their future; health and financial status of family members; high parental expectations; and academic worries, to name some. These stressors are not limited to adolescents with academic difficulty but are faced by typically achieving adolescents as well. Studies point out that at any given point of time 25% students have symptoms of Depression. A Study conducted on students of Punjab University showed overall prevalence of depression, anxiety, stress to be 59.3%, 86.5% and 52.8% respectively. There was high co-morbidity between depression, anxiety and stress (39.5%). There

was concentration of high prevalence of all three disorders in younger age groups. And it was highlighted that there was high co morbidity between Anxieties, depression, stress. These types of symptoms affect their whole life including their studies and academic performance. The study was done to assess the Depression, Anxiety and Stress in relation to academic achievement in students of University. Students at College level have multiple stress in life which manifest in symptoms of DAS affecting significantly their academic achievement. Therefore, this study was planned to find out the pattern of depression, anxiety and stress in relation to academic achievement among government higher secondary school students Chandigarh, city.

Objectives of the study :

- To Assess the Depression, Anxiety & Stress Among Students of higher secondary school.
- To determine the association of Depression, Anxiety & Stress with sociodemographic Variables.

Materials and Methods

A Cross-sectional research design was adopted. Quantitative method of data collection was adopted as a study method. The study area was government higher secondary of selected schools of Chandigarh. The study population for this research study was adolescents of government higher secondary schools of Chandigarh. Probability proportionate Sampling technique was used. (PPS)

Table 1: Sample collection from 5 different government higher secondary school

Government School	Number Of Students	Probability of Students	Sample of students	Actual sample
Halo mazra	50	0.16	46	45
19 Model senior secondary govt. school	59	0.19	55	54
19 Model higher secondary govt. school	70	0.22	63	67
21 Model higher secondary govt. School	57	0.18	52	53
22 Model higher secondary govt. school	70	0.22	63	69
TOTAL	306	0.97	288	288

Sample size-

Taking expected proportion as 50% with precision of 5%, the sample size was calculated as 288 subjects in a finite population of 15,650 individuals.

Inclusion criteria:

- Students Studying In Higher secondary schools of selected government schools of Chandigarh
- Male and Female Students who were willing to participate in the study.
- Those who will be present on the day of survey.

Exclusion criteria:

- Students who were having physical or mental problem.

Data collection tools- Appropriate literature review was done before the conduction of the study. Questionnaire was constructed taking reference of various research papers. Pretesting and modifications of questionnaire was done in every aspect of the study as per expert's opinion.

The study period was from January 2019 to May 2019. Pilot study was conducted on 20 students of higher secondary schools who were included in main study. Approval was taken from the ethical committee of Akal School of Public Health, Eternal University, Baru Sahib, Himachal Pradesh (India). All the respondents were well informed about the objectives and purpose of the study. Permission was also taken from Director of Senior secondary Education, Chandigarh and Principals of the selected schools. Informed consent was obtained from the individual participants. Confidentiality was maintained and the results were used for the research purpose only. Data analysis software SPSS version 23 was used for data entry, processing and analysis. Descriptive and inferential statistics were performed for data analysis.

Results**Table 1: Distribution of students according to sociodemographic variables**

Socio- Demographic Variables		FREQUENCY	PERCENT
AGE (years)	13-15	65	22.6%
	16-18	210	72.9%
	19-21	13	4.5%
GENDER	Male	175	60.8%
	Female	113	39.2%
RELIGION	Hindu	239	83.0%
	Muslim	17	5.9%
	Sikh	26	9.0%
	Christian	6	2.1%
EDUCATION	Primary	2	0.7%
	Secondary	3	1.0%
	Higher Secondary	57	19.8%
	Graduation	255	78.1%
OCCUPATION OF FATHER	Govt. Employee	39	13.5%
	Pvt. Employee	113	39.2%
	Self employed	75	26.0%
	Daily wage	15	5.2%
	Homemaker	1	0.3%
	Agriculturist	7	2.4%
	Retired	3	1.0%
	Other	35	12.2%
OCCUPATION OF MOTHER	Govt .employee	12	4.2%
	Private Employee	24	8.3%
	Self-employed	15	5.2%
	Daily Wage	2	0.7%
	Homemaker	206	71.5%

	Agriculturist	1	0.3%
	Others	28	9.7%
ECONOMIC STATUS	>40430	49	17.0%
	40429-20210	55	19.1%
	20209-15160	51	17.7%
	15159-10110	65	22.6%
	10109-6060	52	18.1%
	6059-2021	13	4.5%
	<2020	3	1.0%
FAMILY TYPE	Nuclear	193	67.0%
	Joint	80	27.8%
	Extended	11	3.8%

Table 1 shows frequency of socio demographic characteristics of the respondents Majority of respondents,(60.8%) were male and (39.2%)female.(22.6%) were in the age group of 13-15 years, (72.9%)were in 16-18years ,(4.5%) were in 19-21 years. (17.0%) of the respondents had income (>40430) per month , (19.1%) of the respondents had income(40429- 20210),(17.7%) have monthly income (20209-15160), 22.6%(15159-10110),18.1% (10109-6060).4.5% (6059-2021), and only1.0% (<2020). Majority of respondents 67.0% belonged to nuclear family type and rest of 27.8% belonged to joint and mere 3.8% belonged to extended family type. Majority of respondent's mother were homemaker i.e. 71.5% and 4.2 %were govt employee whereas 8.3% were private employee and 5.2% were self-employed,0.7% were on daily wages, 0.3% were agriculturalist & 9.7% belonged to other category. Majority of the respondents father occupation was 39.2% that of private employee, 26.0% were self-employed, 13.5% were Govt employee,5.2% were on Daily wages1.0 %were retired and 12.2% % were in other category. Majority of the respondents education is 78.1% were higher secondary ,19.8% were of senior secondary. Majority of respondents 83.0% belonged to Hindu religion, 5.9% were Muslim, 9.0% were Sikh, and only 2.1% were Christian.

Regarding distribution of depression among higher secondary school students that majority of students i.e 35.1 % were normal ,29.9% were moderately depressed,18.8% had mild depression, 13.9 % were severely depressed and 2.4% were extremely depressed. Also, it was seen that majority of students, 26.4% have moderate anxiety, 25.3% were severe anxiety, 21.5% have Extremely severe anxiety and 18.1 % were normal and 8.7% have mild anxiety.

Regarding distribution of stress, 44.8% were normal, 21.5 % of students have mild stress, 25.0 % have moderate stress, 8.3% have severe stress, and 0.3 % were extremely severe stress.

Table 2: Overall Frequency distribution of depression , anxiety and stress

Table 4.7: Overall Frequency distribution of depression , anxiety and stress					
	Levels				
Category	Normal	Mild	Moderate	Severe	Extreme
Depression	35.1%	18.8%	29.9%	13.9%	2.4%
Anxiety	18.1%	8.7%	26.4%	25.3%	21.5%
Stress	44.8%	21.5%	25.0%	8.3%	0.3%

Table 3: Association of Anxiety and Depression

Depression					
Anxiety		No	Yes	P- Value	Odds Ratio
	No	41	11	< 0.001**	10.93
		78.8%	21.2%		
	Yes	60	176		
		25.4%	74.6%		

* Statistically significant at $p < 0.05$ ** Statistically highly significant at $p < 0.05$ This table shows results of association of anxiety and depression is significant i.e $p < 0.001$ & OR=10.93

Table 4: Association of Stress and Depression

Depression					
Stress		No	Yes	P- Value	Odds Ratio
	No	85	44	<0.001**	17.26
		65.9%	34.1%		
	Yes	16	143		
		10.1%	89.9%		

*Statistically significant at $p < 0.05$ **Statistically highly significant at $p < 0.05$ This table show significant association between stress and depression to be < 0.001 & $OR = 17.6$

Table 5: Association of Stress and Depression

Stress					
Anxiety		No	Yes	P- Value	Odds Ratio
	No	44	8	<0.001**	9.771
		84.6%	15.4%		
	Yes	129	159		
		44.8%	55.2%		

This table shows significance between anxiety and stress as $p < 0.001$ and $OR = 9.771$

Table 6: Association of Depression with Socio-Demographic Variables

Gender	Depression					P- Value
	Normal	Mild	Moderate	Severe	Extremely severe	
Male	68	36	51	16	4	0.04*
	38.9%	20.6%	29.1%	9.1%	2.3%	
Female	33	18	35	24	3	
	29.2%	15.9%	31.0%	21.2%	2.7%	

Above Table 4.10 shows significance association of Depression with sociodemographic variables i.e. $p = 0.04$

Table 7: Association of Anxiety with Socio-Demographic Variables

Gender	Anxiety					P- Value
	Normal	Mild	Moderate	Severe	Extremely severe	
Male	42	18	52	46	17	<0.001**
	24.0%	10.3%	29.7%	26.3%	9.7%	
Female	10	7	24	27	45	
	8.8%	6.2%	21.2%	23.9%	39.8%	

Above table 7 shows significance association of anxiety with sociodemographic variables.

Table 8: Association of Anxiety with Socio-Demographic Variables

Gender	Stress					P- Value
	Normal	Mild	Moderate	Severe	Extremely severe	
Male	95	39	32	9	0	<0.001**
	54.3%	22.3%	18.3%	5.1%	0.0%	
Female	34	23	40	15	1	
	30.1%	20.4%	35.4%	13.3%	0.9%	

Table 8 depicts association of Stress with Socio demographic Variables is significant with $p < 0.001$

Table 9: Correlation among Depression, Anxiety and Stress

	Pearson's Correlation (r) Sig (P-value)	Depression	Anxiety	Stress
Depression	r P- Value	1	0.617 (<0.001)**	0.664 (<0.001)**
Anxiety	r P- Value		1	.575 (<0.001)**
Stress	r P- Value			1

Table 9 shows correlation of DASS. It Shows correlation between depression to Anxiety i.e., 0.617 and depression with stress is 0.664. Correlation between anxiety and stress is 0.575 which is indicating significant findings.

Table 10: Comparison of Depression, Anxiety and Stress with Socio-Demographic variables

DAS	Mean	SD	t	Df	Sig(2- tailed)
Depression-Anxiety	-1.03819	5.81737	-3.029	287	0.003**
Depression-Stress	-2.56597	5.29241	-8.228	287	<0.001**
Anxiety-Stress	-1.52778	5.54355	-4.677	287	<0.001**

*Statistically significant at $p < 0.05$ **Statistically highly significant at $p < 0.05$

Table 10 explain the findings of paired t-test. It is depicted that Depression, Anxiety and Stress are significantly associated with each other as association of 0.003 between depression and anxiety, <0.001 between depression and stress, <0.001 between anxiety and stress is seen significant.

Table 11: Comparison of Depression, Anxiety and Stress with Socio-Demographic variables

Socio-Demographic variables	Depression F value, (P- Value)	Anxiety F value, (P- Value)	Stress F value, (P- Value)
Age	3.477, (0.009)*	11.74, (0.00)**	5.15, (0.001)**
Gender	2.53, (0.04)*	-	6.47, (0.00)**
Monthly Income	2.457, (0.046)*	-	-
Occupation of mother	-	2.86, (0.024)*	-

*Statistically significant at $p < 0.05$ **Statistically highly significant at $p < 0.05$

Table 11 shows the significant differences among the levels depression, anxiety and stress among students with respect to their sociodemographic variables.

DISCUSSION

This cross sectional descriptive study was conducted to assess the stress, anxiety and depression among higher secondary school students of Government schools of Chandigarh. The World Health Organization (WHO) reported in 2001 that about 450 million people worldwide suffer from some form of mental disorder and that one in four people meet criteria at some point in their life. Depression is a major cause of morbidity, with a lifetime prevalence varying from 3% in Japan to 17% in the US. In most countries, the number of people who would suffer from depression during their lives falls within an 8-12% range. At any given point of time, 25% students reported symptoms of Depression. In India 90% of people who need mental health support do not have access to it. There is a significant human resource gap of mental health professionals.

According to WHO report, practically all psychiatrists who have had extensive experience in working with school students agree that about 10% of the members of any institution of higher learning are likely to have emotional problems at some time or other during each year which interferes seriously with their work. Very few institutions have sufficiently well-developed psychiatric services to deal with that number of students. Depression, anxiety, stress and substance abuse are mere symptoms of the hidden, unresolved, and ignored emotional issues which cumulatively grow inside insidiously. This study aimed to find the status of the mental health of students of the Punjabi University.

In present study the respondents were between age group of (15-19) and 39.2% of them were female where as 60.8% were male. Majority of the students belonged to age group (16-18) that is 72.9%. Majority of the students population belonged to Hindu religion that is 83.0% among which 97.9 % belonged to higher secondary education category. The findings were found to have similarity with a study conducted in Haryana on school going adolescents which showed mean age 16.94 and SD 0.73 of male group while the mean age and SD for female group was 15.04±0.92. All participants were Hindu, belongs to rural area and studying in 10th, 11th and 12th class. It was seen that female students scored higher than male students in all variables. Differences on all three variables were found to be significant at (0.05) level of significance¹².

This study reveals that out of 288 students population, 35.1 % students were normal whereas rest 29.9% had moderate depression, 13.9 % had severe depression, 18.8 % had mild depression and 2.4 % had extreme severe depression. 18.1 % had no anxiety normal, 26.4% had moderate anxiety, 25.3% had severe anxiety, 21.5% had extreme severe anxiety, 8.7% had mild anxiety. For stress level, it was shown that 44.8% were normal, 25.0 % had moderate stress, 21.5% had mild stress, 8.3% had severe stress, 0.3% had extreme severe stress. Similar results were found by the study conducted in Chandigarh in Punjab University which showed overall prevalence of depression, anxiety and stress to be 59.3%, 86.5% and 52.8% respectively¹³. A similar study conducted by Teh C.K.*et al.* in Malaysia where the prevalence for moderate to extremely severe depression, anxiety and stress are 30.7%, 55.5%, and 16.6% respectively ¹⁴, this is lower than another study done among Malaysian university students whereby the percentages are 37.2%, 63.0%, and 23.7% for depression, anxiety and stress. However, in present study severe and extremely severe anxiety together made 43.4% which was about 13% higher than the study done among Malaysian university students where 29% had severe or extremely severe anxiety. High prevalence of anxiety was due to high number of females presenting severe and extremely severe anxiety. Moreover, the sample had more females than males, which have perhaps exaggerated the prevalence of anxiety. In the present study 16% of males had severe and extremely severe depression whereas females accounted for 9%; which is similar to the findings from study on students of University of Malaysia.

The association within the DAS variables also showed some significant results that is association of depression with anxiety was found to be significant at $p < 0.001$ and odds ratio is 10.93. Association of depression and stress was found to be significant at $p < 0.001$ and odds ratio is 17.26 and for anxiety and stress it was < 0.001 and odds ratio was 9.771. Also the association of depression, anxiety and stress with gender was found to be significant at $p < 0.04$, $p < 0.05$ and $p < 0.001$ respectively.

In present study additional findings were observed using Correlation, paired t-test and ANOVA. Correlation between the scores of Depression and anxiety was found to be 0.62, for depression and stress, it was 0.664, and r value for anxiety and stress was 0.575. Correlation is significant at the < 0.001 level. Paired showed significance of 0.003 between depression and anxiety, < 0.001 between depression and stress, < 0.001 between anxiety and stress for paired test and for linear regression showed significant results 0.013 for age, 0.016 for economic status of the family, 0.005 for family type, 0.00 for anxiety and 0.00 for stress.

In a similar study, findings were almost same. The total number of respondents in the survey was 242. The age of the students ranged from 13-18 years (mean 15 ± 1.1 years). Among the students, 138 (57.0%) were males. The respondents included 47, 70, 64 and 61 from class 9th, 10th, 11th and 12th respectively. A large majority of the students were Hindu (91.7%). Nearly 80% of the students (194) belonged to nuclear families. Correlation between the scores of Depression and anxiety was found to be 0.454, for depression and stress, it was 0.647, and r value for anxiety and stress was 0.586. Correlation is significant at the < 0.001 level. Correlation represent the Pearson's correlation coefficient scores of the students. However, the scores in the three domains namely Depression, Anxiety and Stress were found to be remarkably correlated. It was seen that depression was significantly more among the females (mean rank 132.5) than the males (mean rank 113.2), $p = 0.03$. Anxiety and Stress scores were however not significantly different between the two genders. No significant association was observed with the type of family (nuclear or joint) or mother's education level (graduate or lesser) or income group of the respondent's family¹⁵.

CONCLUSION

Twenty first century can be indicated as the century of competition. Everywhere there is a race, a competition. With the huge development in science and technology, all part of the world is connected with this competition. Especially in the field of education, this competition rapidly increased among the students. To survive in this competition, educational achievement is necessary and compulsory to all and for good educational achievement, anxiety free, a positive environment is also essential. Academic anxiety is a deliberating factor which impact on students & academic achievement. Adolescence is related to considerable psychological distress irrespective of academic performance. Stress is faced by adolescents irrespective of whether they are doing well in academics or not. Typically achieving students experienced

a higher level of stress and suicidal ideas than students having academic difficulty. The problems arise in various areas such as relationship with parents, parental expectations, worrying about future, role as students and peer relationships, and academic difficulties. Thus, assessment of stress among adolescents should extend beyond academic difficulties. Teachers and parents need to be more aware of the multiple issues leading to stress among students to learn ways to handle and guide adolescents.

LIMITATION:

1. The Study involved self-administered method of data collection and hence the limitations pertaining to this method may be present in the study.
2. The study was limited to only students of selected government schools of Chandigarh out of 20 government schools

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